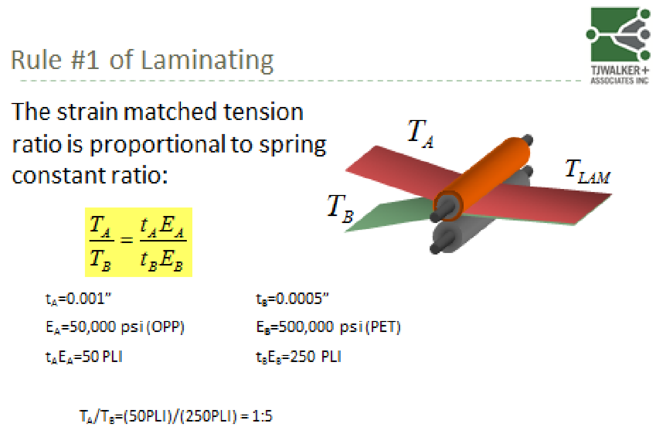
CURL MECHANICS – MACHINE DIRECTION CURL

Curl is common in coated and laminated products. Curl is created when two materials are bonded together at differential strains. Laminates curl toward the side with the shorter dimension.

Machine direction (MD) curl is the easiest to understand. Tension will elongate a web. The percent elastic strain in a web will be proportional to the web tensile stress (force of tension divided by cross-sectional area of thickness times width, stress = F/tw). For polyester webs, MD strain will be a small number – less than one percent. For a typical polyester, the strain will be 0.2% for 1 lb/in tension on a 1 mil web (assumes the elastic modulus is 500,000 psi). Lower tension will have lower MD strain. Thicker webs will have lower MD strain.



Making a flat laminate in in the machine direction starts with MD strain matching. If you laminate two 1-mil polyester webs at 1 PLI tension, each will have 0.2% strain when they are bonded together and each will want to relax 0.2% when tension is removed. If one of the 1-mil web had higher tension, it would have higher strain at bonding, relax more when tension is removed, and the sample would curl toward the side of the higher tension, higher strain web. If 1-mil and 2-mil polyester webs were laminated with each having 1PLI tension, the thinner 1-mil web would have higher strain at bonding, relax more when tension is removed, and the sample would curl toward the 1-mil side. If the 1-mil and 2-mil polyester webs were laminated with 1PLI and 2PLI respectively, then they would be bonded at the same strain, relax the same when tension is removed, and have no MD curl. The 1 and 2-mil webs would also make MD flat laminates at any combination of tensions where the 2-mil tension is two times the 1-mil tension.

CURL MECHANICS – TRANSVERSE DIRECTION CURL

Transverse direction curl is a little more difficult to understand. I can think of three common sources of TD curl: 1) Mismatched MD strains of equal Poisson's ratio materials, 2) materials with different Poisson's ratios, and 3) materials that have other dimensional change mechanisms (broad third category).

Why are MD and TD curl commonly in opposite directions (towards different sides)?

The top cause of TD curl is tension related, but is a secondary effect caused Poission's contraction. When materials elongate with tension they don't necessarily change their density; therefore, any increase in length will be accompanied by a decrease in thickness and width. The thickness change is nearly undetectable, but the width change can be significant. For most solid materials, Poisson's ratio (the negative ratio of one directional strain to another) is 0.3. For example, if a web has 0.2% MD strain and a Poisson's ratio of 0.3, the TD strain will be -0.06%. (This would create a 0.036-in change over 60-inch width.)

In laminating similar materials, the Poisson's ratio explains why most samples will curl toward one side (positive) in the MD and toward the other side (negative) in the TD directions. Also, if MD strains are matched, then TD strains will also be matched and the untensioned sample will be flat in both MD and TD directions.

Why would a sample have TD curl if there is no tension in the crossweb direction?

If materials have unequal Poisson's ratios, then matching MD strain will not match TD strain. Many solid materials, those without a porous structure, will have a typical Poisson's ratio of 0.3. However, material with air voids within their structure, such as porous films or foams, fibrous materials (nonwovens, papers), or textiles, can void their air space during tensioning, changing their density. These type of materials can have Poisson's ratios of one or higher. In general, the greater the air space in the web, the higher the Poisson's ratio can be. This mechanism should not be a factor in your window film products.

CURL MECHANICS – NON-TENSION DIMENSIONAL CHANGE

What else can cause the dimensional change leading to curl?

Usually curl involves a web with layers that have different material properties. In the above discussion, curl is created by the transition of bonding under tension-induced strain then checking for curl when tension is released. However, curl can be created from other sources of dimensional change. Paper and nylon webs will grow and shrink with moisture content. Many coatings will shrink as water or solvents are driven out in a drying process. Most materials will grow and contract with temperature changes. Polymer films may have a stress-heat memory of their previous processing. All of these sources of dimensional change can contribute to curl in coated and laminated products.

Thermal Expansion: A bi-metallic strip (used in old style thermostats) curls with temperature change due to layers with different coefficients of thermal expansion.

Moisture: A single sheet of papers can curl from differential moisture or yielding. Many papers will curl away from moisture, but then curl back toward that side if it is over dried with external heat (commonly referred to as curling to the last side dried). Paper will commonly curl more in the TD direction since the fibers are usually oriented along the MD direction, where moisture makes the individual fibers swell in diameter more than change length.

Yielding: Though paper is not a laminate, it can curl when the fibers of one side of the paper behave differently than the fibers on the other side of sheet. Curled ribbons is made by pulling paper (or woven ribbon) over a sharp edge, yielding one side to be longer than the other.

Density Changes: Many coatings will contract when water or solvents are driven off in a drying process. This may lead to more TD curl since the force of web tension can offset the contracting forces of drying or curing, but without TD forces the web is free to curl in the crossweb direction. Hard coat shrinkage is a strong suspect in Galaxie and Hi Lite laminate shrinkage. This is difficult to assess since the process of applying and curing the hard coating requires heat. Film can be run through the hard coat process step with and without coating, but it may not accurately separate the effects of the coating vs. curing process heat since curing and drying will react differently without the coating present.

Heat Shrinkage:

Polyester (PET) films may expand thermally, but the more dramatic change is usually heat-related shrinkage. In processing PET films, the film is first extruded at ten times the desired thickness then stretched 3:1 in both the MD and TD directions (either sequentially or simultaneously) to reach the final desired thickness. However, the stretching processes can be partially reversed if a film is reheated. Reheating a PET film above a given temperature, such as the glass transition temperature of 170F, allows the film's polymer molecules to move and recoil a small percent from the orientation process, possibly shrinking both the MD and TD directions.

Different polyester film processes will create a film with higher or lower heat shrinkages. Many factors influence heat stability, including tension and temperature in the post-orientation quenching step, post-orientation re-processing (heat relaxing), and heat soaking of wound rolls.

Heat shrinkage is tested with samples heated by air or boiling water and either unrestrained or at a given tension. Unrestrained samples will nearly always shrink with heat. Tensioned samples may shrink, elongate, or have zero dimensional change depending on the balance of shrink forces to tension. Polyester films are usually designed to have low shrinkage, but may also be intentionally designed as high shrink films that will contract 50% with heat. PET expansion is not a likely curl-causing mechanism, but film shrinkage is a strong possibility and will differ greatly with PET suppliers and processes.

CURL MEASUREMENTS

There are three common tests to quantify curl.

Curl Test Method	Comments
1. MD and TD Strip Tests	Narrow strips, typically 1-in by 12-in, are cut running in the machine and transverse directions. The strips are set down on their side, allowing them to curl into an arc or circle. Curl is quantified by the radius or curvature. Positive and negative curl is arbitrarily assigned to orientation of the curl towards the 'top' or 'bottom' side of the web. For coated webs, negative curl is usually towards the coated side.
2. X Test	An X is cut in a sample running 45-degrees from the MD and TD directions. The triangular wings created by the X will curl up or down from the plane of the web (or stay in plane for curl-free samples). Curl can be quantified by using a fixed size or template to create the X and using a ruler to measure the distance from the web plane to the tip of each triangle (you will need to flip the sample over to measure positive curl).
3. Circle Test	The circle test is good for finding the dominant axis of curl. Not all samples will curl primarily along the cardinal headings of MD and TD. When laminating with uneven nip loads, the product may have different axis of curl across the web's width. Like the strip curl test, magnitude of curl measured as the diameter or radius of the arc or circle formed by the sample's curl.

For wide processes (more than 12-in), any curl test should be done across the web's width, at least three positions, such as operator side, web center, and drive side.

Curl magnitude or curvature is a function of three structural factors: thickness of each layer, elastic modulus of each layer, and the strain mismatch (whether from tension, thermal, or moisture changes). Both the strip test and circle test radius or diameter measurements can be converted into strain mismatch if the material thickness and elastic modulus or known. A 0.1% strain mismatch between two layers will create different radius of curl depending on the stiffness of the layers, which is a function of thickness and elastic modulus. Thinner webs are much more prone to small radius curl. A thick web with a dimensionally different, but thin second layer, may resist bending and curvature even under high strain mismatching.